

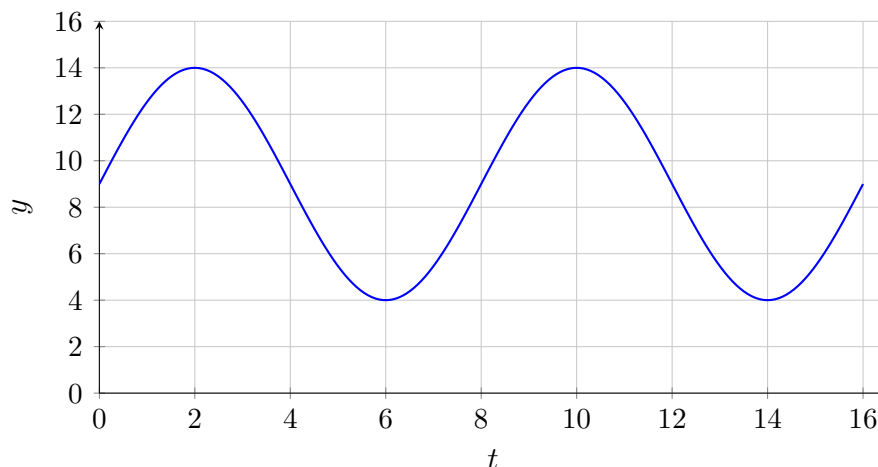
Unit Circle and Trigonometric Graphs

Recovering amplitude, midline and period from a graph, building a sinusoidal model from a situation, evaluating that model, and solving a periodic equation on one cycle

Before you start. For $y = A \sin(B(t - C)) + D$:

- $|A| = \frac{\max - \min}{2}$ and $D = \frac{\max + \min}{2}$ (amplitude and midline).
- The period is $\frac{2\pi}{B}$, so $B = \frac{2\pi}{\text{period}}$. Measure a period between two *successive maxima*. A maximum and the next *minimum* are only **half** a period apart.
- Work in radians. Every model on this sheet uses radian measure, so a calculator left in degree mode will return a plausible but wrong number.
- A periodic equation almost never has one solution on a full cycle. Find them all, then keep the ones inside the interval you were given.

Figure 1 — Curve G. *Used by Problems 1 to 4.*



- 1.** Curve G is the graph of $h(t) = 5 \sin\left(\frac{\pi t}{4}\right) + 9$. Find $h(2)$ exactly, then check your value against the graph.

Answer: _____

2. Use **Figure 1**. Curve G rises to a highest value of 14 and falls to a lowest value of 4. Find its amplitude.

Answer: _____

3. Use **Figure 1**. Find the midline of Curve G — the value of y about which the curve oscillates.

Answer: _____

4. Use **Figure 1**. Curve G reaches successive maxima at $t = 2$ and at $t = 10$.

(a) What is the period of Curve G?

Answer: _____

(b) Find the value of B for which Curve G can be written as $y = A \sin(Bt) + D$.

Answer: _____

5. A Ferris wheel has diameter 40 metres. Its lowest point is 3 metres above the ground, and it completes one full turn every 12 minutes. A rider boards at the lowest point at time $t = 0$.

(a) What is the amplitude of the rider's height model, in metres?

Answer: _____

(b) What value of B belongs in $h(t) = -A \cos(Bt) + D$?

Answer: _____

6. The Ferris wheel of Problem 5 is modelled by $h(t) = -20 \cos\left(\frac{\pi t}{6}\right) + 23$, with h in metres and t in minutes. Find the rider's height four minutes after boarding.

Answer: _____

7. The rider is level with the centre of that wheel when $h(t) = 23$, which happens exactly when $\cos\left(\frac{\pi t}{6}\right) = 0$. Find *all* times t in one full turn, $0 \leq t < 12$ minutes, at which this happens.

Answer: _____

8. In a harbour the water is deepest at 12 feet, at $t = 3$ hours, and shallowest at 4 feet, at $t = 9$ hours. The depth is modelled by $d(t) = A \cos(B(t - 3)) + D$.

(a) Find A .

Answer: _____

(b) Find D .

Answer: _____

(c) Find B . (Careful: a high tide and the *next* low tide are half a period apart.)

Answer: _____

9. A different sinusoid reaches a maximum of 18 at $x = 1$ and its next minimum of 2 at $x = 7$.

(a) Find its amplitude.

Answer: _____

(b) Find its midline.

Answer: _____

(c) Find its period.

Answer: _____

10. Using the harbour model from Problem 8, which is $d(t) = 4 \cos\left(\frac{\pi(t-3)}{6}\right) + 8$, find the depth of the water at $t = 5$ hours.

Answer: _____

11. Find *all* times t with $0 \leq t < 12$ hours at which the harbour depth $d(t) = 4 \cos\left(\frac{\pi(t-3)}{6}\right) + 8$ equals 10 feet. There is more than one; explain briefly how the graph tells you how many to expect.

Answer: _____

12. Synthesis. Two students model the same Ferris wheel from Problem 5. Ana writes $h(t) = -20 \cos\left(\frac{\pi t}{6}\right) + 23$; Ben writes $h(t) = 20 \sin\left(\frac{\pi(t-3)}{6}\right) + 23$.

(a) Evaluate *both* models at $t = 5$, exactly.

Answer: _____

(b) Explain why both students can be right, and say which parameters of a sinusoid a graph pins down uniquely and which it does not.